

# **Notes on Tidal Engineering**

Three Studies of the Lower Channel

by E. R. Halloway

THE COASTAL PRESS

## 1. The Lag of the Channel

Every estuary keeps its own clock. The tide that reaches the quay at noon has already turned twice in the deep channel, and the difference between those two moments is the whole of the engineer's problem.

Measurements taken in the winter of that year showed a consistent lag of forty minutes, which the committee attributed to silting, though no survey had been carried out since the previous decade. Measurements taken in the winter of that year showed a consistent lag of forty minutes, which the committee attributed to silting, though no survey had been carried out since the previous decade.

The commissioners were unwilling to fund a second hydro-graphic survey before the spring, and the matter was deferred.

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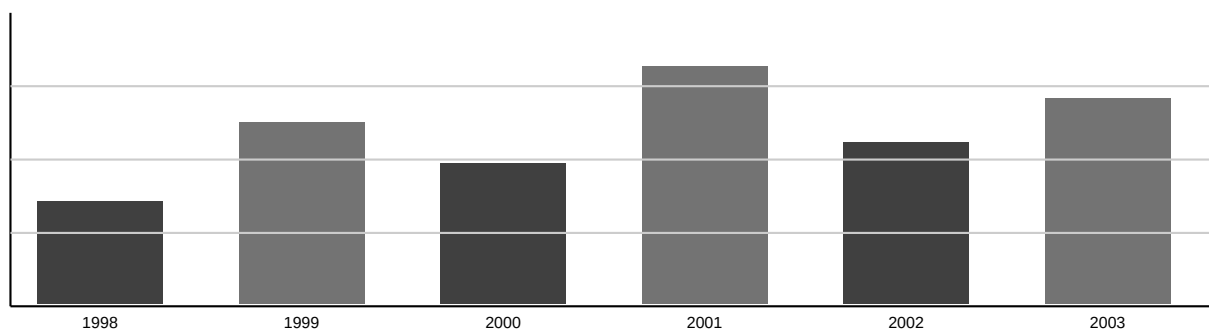
1. Committee minutes, 14 March. The figure is disputed in the appendix.

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#### **OBSERVED DEVIATIONS**

1. forty minutes at the quay
2. twelve minutes in the deep channel
3. no measurable change at the bar

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*Figure 1: Recorded lag at the quay, by year.*

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## 2. Silt and Its Measurement

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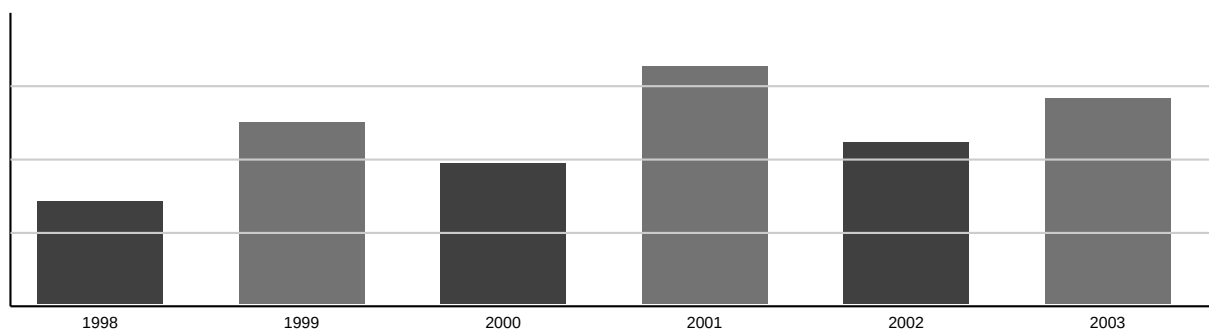


Figure 2: Recorded lag at the quay, by year.

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### 3. A Proposal

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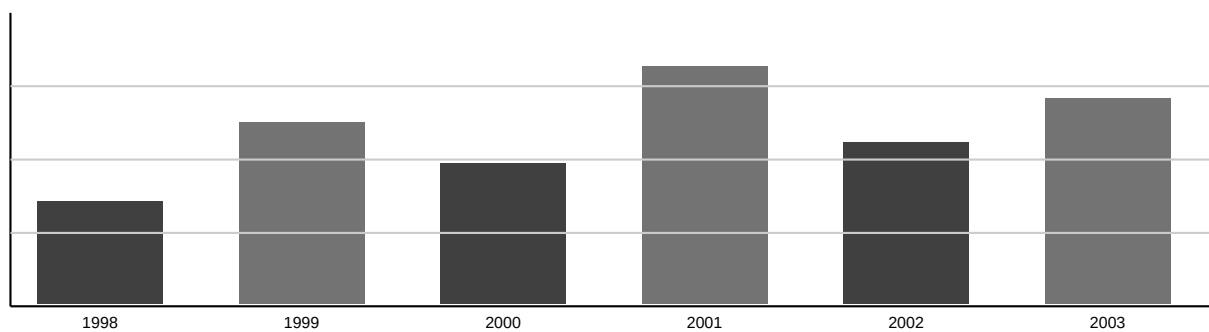


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